

NICHOLAS EARL

📍 Chicago, IL 📞 (517) 402-4590 ✉ nmearl2@illinois.edu 🌐 nmearl 🔗 nicholasearl.me

PROFESSIONAL SUMMARY

I am a research software engineer and astrophysicist whose career has centered on building and sustaining open-source tools for the astronomy community. My astrophysical research focuses on tidal disruption events and nuclear variability, where I use probabilistic modeling and machine learning to study black hole physics and galaxy evolution at survey scale.

EDUCATION

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN | *Ph.D. in Astronomy* Aug 2020 – Present

- Advisor: *K. Decker French*
- Thesis: *Accretion, Transients, and Galaxy Evolution in the Time-Domain Survey Era*

SAN DIEGO STATE UNIVERSITY | *M.S. in Astronomy* Aug 2012 – May 2014

- Advisor: *Jerome Orosz*
- Thesis: *Photodynamical Modeling of Hierarchical Stellar System KOI-126*

MICHIGAN STATE UNIVERSITY | *B.S. in Astrophysics* Aug 2007 – May 2011

- Honors College
- Advisor: *Brian O'Shea*
- Thesis: *Dynamic Evolution and Metal Mixing in Supernova Shock Remnants*

MICHIGAN STATE UNIVERSITY | *B.A. in Philosophy* Aug 2007 – May 2011

- Honors College
- Advisor: *Debra Nails*
- Thesis: *Supervenience for an Infinity of Minds*

PROFESSIONAL EXPERIENCE

HARVARD-SMITHSONIAN CENTER FOR ASTROPHYSICS

Software Engineer & Systems Architect | *NASA/NSF-supported projects* Jun 2021 – Present

- Architect and develop Cosmic Data Stories (CosmicDS) under NASA Science Activation (SciAct) grant, an interactive STEM curriculum platform built within the Glue ecosystem, enabling scientists to create linked-view, data-driven narratives without web-development expertise.
- Lead design of scalable pipelines for story authoring (Jupyter + web components), supporting adoption across middle school, high school, and introductory college courses.
- Architect and maintain CosmicDS deployment and operations infrastructure, including AWS services, Docker container management, database setup, API access control, and production configuration.
- Contribute directly to Glue and Jupyter interoperability layers, improving linked-view visualization and large dataset responsiveness.
- Collaborate with NASA, NSF, and CfA teams on learning goals, accessibility, and open-science reproducibility across CosmicDS modules.
- Through CfA, contribute to joint Earth-2 / NVIDIA efforts on GPU-accelerated climate and astrophysics workflows; explore real-time data federation, interactive visualization, and AI-assisted surrogate modeling for scientific simulations.
- Participate in the development of LIVE Environments, integrating Earth-2-style GPU pipelines into immersive astronomical and Earth-science data exploration tools.

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Urbana, IL

Graduate Research Assistant | *Ph.D. Research*

Jan 2021 – Present

- Investigate nuclear transients, TDEs, and AGN using multi-wavelength photometry and spectroscopy; connect transient behavior to host-galaxy demographics and SMBH accretion physics.
- Led multi-wavelength analysis of AT 2020nov, identifying disk reprocessing signatures and late-time X-ray behavior.
- Developed FEADME, a GPU-accelerated Bayesian modeling engine (JAX/NumPyro) for relativistic elliptical accretion-disk profiles; applied to large AGN/TDE samples to contrast disk geometry and evolution.
- Developed NAPTITUDE, a neural-process framework for Rubin/LSST-scale early-time transient classification, incorporating calibrated uncertainties and irregularly sampled context/target splits.
- Lead EDGE-CALIFA angular-momentum analysis combining eCALIFA stellar kinematics with integrated and resolved CO products to study molecular gas content, structure, and kinematic alignment across the galaxy angular-momentum sequence.
- Design and develop a transformer-based transient classification system to distinguish AGN variability from TDE behavior by integrating multi-wavelength context, stochastic variability, and sequence modeling.
- Active member of the Young Supernova Experiment (YSE) and the LSST TVS TDE subgroup, contributing to transient discovery, simulation, and Rubin-era follow-up strategies.
- Participate in YSE triaging of newly observed fields to identify early supernovae, nuclear flares, and TDE candidates; contribute to early-alert vetting, prioritization, and rapid follow-up coordination.
- Develop infrastructure and catalog integration tools for OTTER (Open multiWavelength Transient Event Repository).

Graduate Teaching Associate

Aug 2020 – Jan 2021

- Taught introductory astronomy concepts to undergraduate students through virtual learning labs.
- Ran office hours, performed grading duties, and provided additional learning spaces for students.

SPACE TELESCOPE SCIENCE INSTITUTE

Baltimore, MD

Senior Software Engineer

Jan 2018 – Aug 2020

- Chaired Spectroscopic Coordination Committee; advanced interoperability across Astropy and next-generation analysis tools, including Glue-Viz, for JWST data analysis.
- Served as an Astropy Core Developer and package lead for Astropy's spectroscopic tools, including Specutils; guided API design, spectral data model development, and community-facing analysis infrastructure; coordinated releases, maintained pytest test suites, managed CI pipelines via GitHub Actions, and contributed Sphinx documentation across the Astropy ecosystem.
- Led cross-instrument, cross-observatory software delivery for JWST science pipelines; collaborated with community developers to integrate STScI workflows into open-source tools.
- Drove notebook-native user tools (Glue/Jupyter) to broaden access to spectroscopic analysis; organized technical colloquia and community hack days.
- Diagnosing the Multiphase CGM (PI): Led development of Spectacle, an automated kinematic line-analysis package for synthetic and observed absorption spectra; quantified resolution-dependent CGM kinematics and compared simulations with HST/COS observations.
- Small-scale Structure of the CGM (Co-I): Ran high-resolution galaxy simulations to resolve sub-kpc CGM structure, modeling mixing, cooling, and absorber formation consistent with COS constraints.
- Grizli / WFC3/IR Spectroscopy: Reduced slitless spectroscopy for the full G102+G141 archive and delivered uniform redshifts and emission-line fluxes; supported dust/attenuation studies and luminosity-function analyses for HST and JWST preparatory science.

Senior Research & Instrument Analyst

Jun 2014 – Jan 2018

- Led teams to deliver cross-instrument spectroscopic tools; coordinated reference-file production for MIRI; verified algorithms with instrument teams.
- Contributed to JWST Exposure Time Calculator (ETC) and collaborative web application.

- Developed the JWST ETC and shipped collaborative analysis features; led 3–5-person coding sprints for astronomy software delivery.
- Built interactive tools for spectroscopic exploration; coordinated MIRI reference files and validation workflows.
- Led development of Spectacle, a synthetic/observational line-analysis package for CGM studies.
- Quantified resolution-dependent CGM kinematics in hydro simulations; compared simulated and observed absorbers to characterize multiphase structure.

SAN DIEGO STATE UNIVERSITY

San Diego, CA

Graduate Research Assistant

Aug 2012 – May 2014

- Worked with Dr. Jerome Orosz and Dr. William Welsh on master's thesis research characterizing compact multi-body stellar and exoplanetary systems identified in the Kepler telescope field.
- Developed and extended ELC and Pynamic modeling tools for photodynamical analysis of eclipsing hierarchical systems, combining orbital dynamics, light-curve modeling, and parameter inference.
- Built computational analysis programs to process Kepler photometry, identify instrumental and astrophysical systematics, and refine physical parameters for multi-body systems.

Graduate Teaching Associate

Aug 2012 – May 2013

- Taught undergraduate astronomy laboratory courses for Dr. Orosz and Dr. Sandquist, including lecturing, hands-on experimentation, and trips to SDSU's Mount Laguna Observatory.

GRANTS, AWARDS, & FELLOWSHIPS

2025 First Author Publication Award, University of Illinois Urbana-Champaign

2021, 2023 Center for AstroPhysical Surveys Fellowship, University of Illinois Urbana-Champaign.

2018 STScI Achievement Award – Spectroscopy Visualization Tools, Space Telescope Science Institute.

2017 STScI Achievement Award – JWST ETC for CfP, Space Telescope Science Institute.

2017 Resolving the Small-scale Structure of the CGM, HST Program AR-15012 (Co-I).

2016 Diagnosing the Multiphase CGM, HST Program AR-14560 (PI).

CONFERENCE & INVITED TALKS

AT2020nov: Rare Transient Event in a Pre-Existing Accretion Disk

- Boom! Transient Workshop at UIUC, Champaign, IL, 2022.
- Transient and Variable Universe Conference at UIUC, Champaign, IL, 2023.
- The Restless Nature of AGN, Naples, Italy, 2023.

Photospheric Disk Modeling with Multi-Wavelength Photometry

- Anticipating the Rising Tide of Tidal Disruption Events Conference at KITP, Santa Barbara, CA, 2024.

Bayesian Modeling for Accretion Disk Analysis

- Invited Talk at UC Santa Cruz, CA, 2024.

Neural Processes for TDE Identification & Forecasting

- Invited Talk at NCSA Center for AstroPhysics, UIUC, Champaign, IL, 2024.
- SkAI Hackathon at SkAI Institute, Chicago, IL, 2026.

GPU-Accelerated Bayesian Modeling for Accretion Disks using JAX

- X-ray Quasi-Periodic Eruptions & Repeating Nuclear Transients Conference at the European Space Astronomy Centre (ESAC) in Madrid, Spain, 2025.
- European Astronomical Society conference in Cork, Ireland, 2025.

Mapping Molecular Gas Bimodalities in the Angular Momentum Plane with EDGE-CALIFA

- EDGE Splinter Workshop at UIUC, Champaign, IL, 2026.

MENTORING & OUTREACH ACTIVITIES

Mentoring

- **Society for Equity in Astronomy Mentorship Program:** Mentored two astronomy undergraduates through UIUC's Society for Equity in Astronomy mentorship program, supporting academic planning, research preparation, and professional development.
- **Warrior-Scholar Project at UIUC:** Served twice as Project Lead for UIUC Warrior-Scholar Project programming, supporting academic bridge experiences for student veterans entering or returning to higher education.

Activities

- **UIUC Society for Equity in Astronomy:** Secretary for graduate-student-led astronomy equity group supporting inclusive departmental programming and community-building.
- **2022 UIUC AstroFest:** Organizer for annual astronomy research event featuring talks, posters, and department-wide scientific exchange.
- **Astropy Coordination Conference 2019:** Scientific organizing committee member for community meeting supporting coordination across the Astropy open-source astronomy software ecosystem.
- **Python in Astronomy 2019:** Scientific organizing committee member for conference bringing together astronomy Python developers, educators, and contributors through talks, tutorials, unconference sessions, and sprints.
- **San Diego State's Schwartz Astronomical Society:** Council member of SDSU astronomy student society supporting student community, rooftop observing, and astronomy events.
- **Society of Physics Students at Michigan State University:** Member of MSU chapter connecting physics and astronomy students through talks, career preparation, research discussions, and student community.

Outreach

- **Sexual-Orientation and Gender Minorities in Astronomy (SGMA):** Member of AAS committee supporting equality, visibility, and professional inclusion for sexual-orientation and gender minorities in astronomy.
- **OUT in Science, Technology, Engineering, and Mathematics (oSTEM):** Member of national LGBTQ+ STEM organization supporting community building, professional development, and educational programming.
- **The Alliance LGBT Group at Michigan State University:** Member of MSU student organization advocating for queer and allied students and supporting campus LGBTQ+ community.
- **MLO Summer Visitor's Program (SDSU):** Organizer for Mount Laguna Observatory public visitor programming connecting the San Diego community with telescope observing and astronomy education.
- **STEM Exploration Day (SDSU):** Organizer for hands-on STEM outreach introducing students and community participants to science activities and pathways.
- **Little Einsteins (SDSU):** Organizer for youth-focused science outreach introducing astronomy and STEM concepts to younger learners.
- **Astronomy Nights on the Roof (SDSU):** Organizer for rooftop observing events with telescopes, informal instruction, and public engagement.
- **Sidewalk Astronomy (STSci):** Organizer for accessible public observing sessions bringing telescopes and astronomy conversations into shared community spaces.

TECHNICAL SKILLS

Scientific Programming Python (primary), NumPy, SciPy, Matplotlib, C/C++, Rust, FORTRAN; additional experience with C#, Julia, MATLAB, and IDL.

Machine Learning & Inference JAX, NumPyro, PyTorch; Bayesian inference, Gaussian and Neural Processes, variational inference, time-series and spectral modeling; uncertainty quantification.

High-Performance Computing GPU acceleration, JIT compilation, parallelization; experience with large simulation datasets and survey-scale pipelines.

Data & Visualization Astropy ecosystem (Specutils, JDAViz), Glue/Glue-Viz, Jupyter; FITS, HDF5, CSV, and survey data products; interactive and linked-view visualization.

Software Engineering Practices Git/GitHub, GitHub Actions, pytest, Sphinx, CI/CD, code review, issue triage, release coordination, and user documentation.

Web & Infrastructure AWS, JavaScript/TypeScript, React/Vue; scientific web applications and cloud-deployed analysis tools.

Platforms Linux/Unix, macOS, Windows.

SELECT OPEN-SOURCE PROJECTS

- **[Lead Developer] NAPTIME (Neural Astrophysical Photometric Transient Identification and Modeling Engine):** Neural-process framework for Rubin-era photometric transient classification and TDE-focused candidate recovery.
- **[Lead Developer] FEADME (Fast Elliptical Accretion Disk Modeling Engine):** GPU-accelerated Bayesian fitting framework for modeling double-peaked and broad-line AGN/TDE spectra.
- **[Lead Developer] Cosmic Data Stories (CosmicDS):** NASA-/NSF-supported interactive data-story platform for STEM classrooms, built within the Glue-Viz ecosystem.
- **[Lead Developer] ipywwt / solara-state:** Supporting packages for interactive scientific web applications, including WorldWide Telescope integration in Jupyter and reusable state-management infrastructure for Solara-based tools.
- **[Developer] Glue / Glue-Viz:** Linked-data visualization framework for multidimensional datasets; contributed to core and web/Jupyter interoperability layers.
- **[Developer] OTTER (Open multiwavelength Transient Event Repository):** Public repository and analysis platform for multiwavelength transient events; developed infrastructure and catalog integration.
- **[Former Lead Developer] JDAViz / SpecViz / CubeViz / MOSViz:** Next-generation visualization and analysis tools in the Astropy ecosystem.
- **[Former Lead Developer] Specutils:** Core spectral analysis library in the Astropy ecosystem; contributed models, algorithms, and infrastructure for 1D spectral manipulation and calibration.
- **[Developer] Astropy:** Contributor to the Astropy project, including general and spectroscopic infrastructure, supporting community-wide standards for astronomical analysis.
- **[Developer] Pandeia (JWST ETC):** Analytical engine and collaborative web app for JWST exposure time calculation.
- **[Lead Developer] Spectacle:** Line analysis framework for synthetic and observational spectra.
- **[Lead Developer] Pynamic / ELC:** Photodynamical and binary-modeling toolsets.

PUBLICATIONS

PAPERS IN PROGRESS

1. **Earl, N.**, et al., in prep., NAPTIME: A Neural-Process Framework for Family-Level Rubin Alert Classification with a Tidal Disruption Event Focus.
2. **Earl, N.**, EDGE-CALIFA Collaboration, et al., in prep., Stellar Angular Momentum and Molecular Gas Across the EDGE-CALIFA Sample.

PREPRINTS

1. Shepherd, M., et al., incl. **Earl, N.**, 2026, The Delay Time Distribution of Tidal Disruption Events, *arXiv e-prints*, doi:10.48550/arXiv.2604.04831.
2. Patra, K., Liepold, E., **Earl, N.**, et al., 2026, JWST and Keck observations of the off-nuclear tidal disruption event TDE 2025abcr: An evolving reprocessing layer, *arXiv e-prints*, arXiv:2604.16093.

3. Verrico, M., et al., incl. **Earl, N.**, 2026, Identifying Changing-Look AGN Transitions in Light Curve Data with the Zwicky Transient Facility, *arXiv e-prints*, doi:10.48550/arXiv.2604.12109.
4. **Ear, N.**, et al., 2025, FEADME: Fast Elliptical Accretion Disk Modeling Engine, *arXiv e-prints*, 10.48550/arXiv.2512.10228.
5. Patra, K., Foley, R., **Earl, N.**, et al., 2025, JWST and Keck Observations of the Off-Nuclear TDE AT 2024tvd: A Massive Nuclear Star Cluster and Minor-Merger Origin for its Black Hole, *arXiv e-prints*, doi:10.48550/arXiv.2510.12572.
6. Jacobson-Galan, W., et al., incl. **Earl, N.**, 2025, An Updated Detection Pipeline for Precursor Emission in Type II Supernova 2020tlf, *arXiv e-prints*, doi:10.48550/arXiv.2501.08475.

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1. French, K., et al., incl. **Earl, N.**, 2026, Prospects for Measuring Black Hole Masses using TDEs with the Vera C. Rubin Observatory, *Publications of the Astronomical Society of the Pacific*, 138, 014101, doi:10.1088/1538-3873/ae2cca.
2. Franz, N., et al., incl. **Earl, N.**, 2026, The Open multiwavelength Transient Event Repository (OTTER): Infrastructure Release and Tidal Disruption Event Catalog, *The Astrophysical Journal*, 999, 243, doi:10.3847/1538-4357/ae346e.
3. Izzo, L., et al., incl. **Earl, N.**, 2026, Normal or transitional? The evolution and properties of two type Ia supernovae in the Virgo cluster, *Astronomy & Astrophysics*, 706, A381, doi:10.1051/0004-6361/202556425.
4. French, K., et al., incl. **Earl, N.**, 2025, Radio Variability in Recently Quenched Galaxies: The Impact of Tidal Disruption Event or Active Galactic Nucleus-Driven Outflows, *The Astrophysical Journal*, 992, 123, doi:10.3847/1538-4357/adff7a.
5. Jacobson-Galan, W., et al., incl. **Earl, N.**, 2025, Final Moments. III. Explosion Properties and Progenitor Constraints of CSM-interacting Type II Supernovae, *The Astrophysical Journal*, 992, 100, doi:10.3847/1538-4357/adfa23.
6. Hayes, E., et al., incl. **Earl, N.**, 2025, Characterizing the standardization properties of type Ia supernovae in the z band with hierarchical Bayesian modelling, *Monthly Notices of the Royal Astronomical Society*, 541, 1948–1968, doi:10.1093/mnras/staf1056.
7. Messick, A., et al., incl. **Earl, N.**, 2025, A Large-scale Search for Photometrically Variable Active Galactic Nuclei in Dwarf Galaxies Using the Young Supernova Experiment, *The Astrophysical Journal*, 985, 223, doi:10.3847/1538-4357/adcdff.
8. **Earl, N.**, et al., 2025, AT 2020nov: Evidence for Disk Reprocessing in a Rare Tidal Disruption Event, *The Astrophysical Journal*, 983, 28, doi:10.3847/1538-4357/ad974.
9. Jacobson-Galan, W., et al., incl. **Earl, N.**, 2025, An Updated Detection Pipeline for Precursor Emission in Type II Supernova 2020tlf, *Research Notes of the American Astronomical Society*, 9, 5, doi:10.3847/2515-5172/ada367.
10. Jacobson-Galan, W., et al., incl. **Earl, N.**, 2024, Final Moments. II. Observational Properties and Physical Modeling of Circumstellar-material-interacting Type II Supernovae, *The Astrophysical Journal*, 970, 189, doi:10.3847/1538-4357/ad4a2a.
11. Wang, Q., et al., incl. **Earl, N.**, 2024, A low-mass helium star progenitor model for the Type Ibn SN 2020nxt, *Monthly Notices of the Royal Astronomical Society*, 530, 3906–3923, doi:10.1093/mnras/stae1038.
12. Ward, S., et al., incl. **Earl, N.**, 2023, Relative Intrinsic Scatter in Hierarchical Type Ia Supernova Sibling Analyses: Application to SNe 2021hpr, 1997bq, and 2008fv in NGC 3147, *The Astrophysical Journal*, 956, 111, doi:10.3847/1538-4357/acf7bb.
13. Davis, K., et al., incl. **Earl, N.**, 2023, SN 2022ann: a Type Icn supernova from a dwarf galaxy that reveals helium in its circumstellar environment, *Monthly Notices of the Royal Astronomical Society*, 523, 2530–2550, doi:10.1093/mnras/stad1433.
14. French, K., **Earl, N.**, et al., 2023, Fading AGNs in Poststarburst Galaxies, *The Astrophysical Journal*, 950, 153, doi:10.3847/1538-4357/acd249.

15. Gardner, J., et al., incl. **Earl, N.**, 2023, The James Webb Space Telescope Mission, *Publications of the Astronomical Society of the Pacific*, 135, 068001, doi:10.1088/1538-3873/acd1b5.
16. Aleo, P., et al., incl. **Earl, N.**, 2023, The Young Supernova Experiment Data Release 1 (YSE DR1): Light Curves and Photometric Classification of 1975 Supernovae, *The Astrophysical Journal Supplement Series*, 266, 9, doi:10.3847/1538-4365/acbfba.
17. Angus, C., et al., incl. **Earl, N.**, 2022, A fast-rising tidal disruption event from a candidate intermediate-mass black hole, *Nature Astronomy*, 6, 1452–1463, doi:10.1038/s41550-022-01811-y.
18. Astropy Collaboration, et al., incl. **Earl, N.**, 2022, The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package, *The Astrophysical Journal*, 935, 167, doi:10.3847/1538-4357/ac7c74.
19. Rest, A., et al., incl. **Earl, N.**, 2022, The DECam Young Supernova Experiment, *Transient Name Server AstroNote*, 24, 1.
20. Jacobson-Galan, W., et al., incl. **Earl, N.**, 2022, Final Moments. I. Precursor Emission, Envelope Inflation, and Enhanced Mass Loss Preceding the Luminous Type II Supernova 2020tlf, *The Astrophysical Journal*, 924, 15, doi:10.3847/1538-4357/ac3f3a.
21. Peebles, M., et al., incl. **Earl, N.**, 2019, Figuring Out Gas & Galaxies in Enzo (FOGGIE). I. Resolving Simulated Circumgalactic Absorption at $2 \leq z \leq 2.5$, *The Astrophysical Journal*, 873, 129, doi:10.3847/1538-4357/ab0654.
22. Astropy Collaboration, et al., incl. **Earl, N.**, 2018, The Astropy Project: Building an Open-science Project and Status of the v2.0 Core Package, *The Astronomical Journal*, 156, 123, doi:10.3847/1538-3881/aabc4f.
23. Batalha, N., et al., incl. **Earl, N.**, 2017, PandExo: A Community Tool for Transiting Exoplanet Science with JWST & HST, *Publications of the Astronomical Society of the Pacific*, 129, 064501, doi:10.1088/1538-3873/aa65b0.
24. Pontoppidan, K., et al., incl. **Earl, N.**, 2016, Pandeia: a multi-mission exposure time calculator for JWST and WFIRST, *Observatory Operations: Strategies, Processes, and Systems VI*, 9910, 991016, doi:10.1117/12.2231768.

SOFTWARE

1. Jdaviz Developers, et al., incl. **Earl, N.**, 2025, Jdaviz, Zenodo, doi:10.5281/zenodo.17211099.
2. **Earl, N.**, et al., 2025, astropy/specutils: v2.1.0, Zenodo, doi:10.5281/zenodo.16615456.
3. Jdaviz Developers, et al., incl. **Earl, N.**, 2023, Jdaviz: JWST astronomical data analysis tools in the Jupyter platform, *Astrophysics Source Code Library*, ascl:2307.001.
4. Dencheva, N., et al., incl. **Earl, N.**, 2021, spacetelescope/gwcs: GWCS 0.18.0, Zenodo, doi:10.5281/zenodo.5800080.
5. Boyd, B., et al., incl. **Earl, N.**, 2020, SALSA: A Python Package for Constructing Synthetic Quasar Absorption Line Catalogs from Astrophysical Hydrodynamic Simulations, *The Journal of Open Source Software*, 5, 2581, doi:10.21105/joss.02581.
6. Ginsburg, A., et al., incl. **Earl, N.**, 2019, radio-astro-tools/spectral-cube: Release v0.4.5, Zenodo, doi:10.5281/zenodo.3558614.
7. Batalha, N., et al., incl. **Earl, N.**, 2019, PandExo: Instrument simulations for exoplanet observation planning, *Astrophysics Source Code Library*, ascl:1906.016.
8. Newville, M., et al., incl. **Earl, N.**, 2016, lmfit-py: release 0.9.3, Zenodo, doi:10.5281/zenodo.49428.

ABSTRACTS & THESES

1. Udomprasert, P., et al., incl. **Earl, N.**, 2025, Cosmic Data Stories: Online Interactive Resources for Engaging the Public with NASA Data, American Astronomical Society Meeting Abstracts #245, 245, 318.02.
2. Harrington, J., et al., incl. **Earl, N.**, 2019, Support the Python Numerical Core, *Bulletin of the American Astronomical Society*, 51, 265.
3. **Earl, N.**, et al., 2018, Spectacle and SpecViz: New Spectral Analysis and Visualization Tools, American Astronomical Society Meeting Abstracts #231, 231, 150.20.
4. **Earl, N.**, et al., 2016, SpecViz: Interactive Spectral Data Analysis, American Astronomical Society Meeting Abstracts #228, 228, 317.05.
5. **Earl, N.**, et al., 2016, Diagnosing the Multiphase Circumgalactic Medium, HST Program AR-14560.
6. **Earl, N.**, 2015, Photodynamical modeling of hierarchical stellar system KOI-126, Master's thesis.
7. **Earl, N.**, et al., 2014, Analysis of Refined Parameters of the Eclipsing Hierarchical Triple Stellar System KOI-126, American Astronomical Society Meeting Abstracts #223, 223, 155.21.